CodeSage: Predicting Defect-Prone Software Modules Using Static Code Metrics

Ebrima Gajaga
Bachelor of Computer Applications
The Maharaja Sayajirao University of Baroda
Vadodara, India
Email: gajaga.dev@gmail.com
PRN: 8023058464

Pritibala Patel
Assistant Professor
Department of Computer Applications
The Maharaja Sayajirao University of Baroda
Email: preeti.patel-compapp@msubaroda.ac.in

Fig. 1. QR Code verifying student identity (MSU Baroda)

***Abstract*—Software defects remain a major challenge in large-scale systems, increasing maintenance cost and reducing reliability. This work investigates whether defect-prone code modules can be predicted using static code metrics derived directly from source code.**

We present CodeSage, a system that performs automated code analysis and predicts bug risk at the file level. A dataset is constructed from open-source Python repositories, where labels are approximated using heuristic rules based on structural complexity.

We extract features such as lines of code, cyclomatic complexity, nesting depth, and function-level characteristics. Logistic Regression and Random Forest models are evaluated.

Results show that Random Forest significantly outperforms Logistic Regression, achieving an F1-score of 0.86 and ROC-AUC of 0.90. Cross-validation confirms good generalization without severe overfitting.

These findings demonstrate that static code metrics alone provide meaningful signals for defect prediction.

***Index Terms*—Defect Prediction, Static Code Analysis, Machine Learning, Random Forest, Software Engineering**

I. Introduction

Modern software systems are increasingly complex, making defect detection and prevention challenging. Traditional approaches such as manual code review and testing are time-consuming and often insufficient.

Prior studies indicate that defect-prone modules exhibit measurable characteristics such as:

- High cyclomatic complexity
- Deep nesting structures
- Large code size

Research Question: Can defect-prone modules be predicted using only static code metrics?

II. Problem Formulation

Let:

$$F = \{f_1, f_2, ..., f_n\} \tag{1}$$

where F denotes the set of all source code files in the dataset, and f_i represents the ith file.

Each file is represented as a feature vector:

$$x_i \in \mathbb{R}^d \tag{2}$$

where x_i is a d-dimensional numerical representation of file f_i, and each dimension corresponds to a software metric.

We aim to learn a binary classification function:

$$g : \mathbb{R}^d \to \{0, 1\} \tag{3}$$

such that:

$$g(x_i) = \begin{cases} 1 & \text{if } f_i \text{ is predicted to be defect-prone} \\ 0 & \text{otherwise} \end{cases} \tag{4}$$

where:

- 1 indicates a high-risk (defect-prone) file
- 0 indicates a low-risk (non-defective) file

—

III. Dataset Construction

A. Repository Mining

Python repositories are collected using the GitHub API. Each repository is cloned and parsed locally to extract structural and complexity-based features.

—

B. Heuristic Labeling

Since ground-truth defect labels are not available, we construct proxy labels using a weighted risk function:

$$R(f_i) = \alpha_1 CC_i + \alpha_2 ND_i + \alpha_3 AFL_i + \alpha_4 LOC_i \tag{5}$$

where:

- CC_i = Cyclomatic Complexity of file f_i (measures number of independent execution paths)
- ND_i = Maximum Nesting Depth (deeply nested logic increases complexity)
- AFL_i = Average Function Length (average number of lines per function)
- LOC_i = Lines of Code (size of the file)
- $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ = learned or manually assigned weighting coefficients

The final binary label is assigned using a threshold τ:

$$y_i = \begin{cases} 1 & \text{if } R(f_i) \geq \tau \\ 0 & \text{otherwise} \end{cases} \tag{6}$$

where:

- $y_i = 1$ indicates a defect-prone file
- τ is a decision threshold controlling label sensitivity

Note: These labels are heuristic approximations and do not represent actual runtime bugs or confirmed defects.

—

C. Feature Extraction

Each file f_i is transformed into a feature vector using static code analysis.

Structural Features:

- $LOC(f_i)$: Total number of non-empty lines in file f_i
- $N_f(f_i)$: Number of functions defined in f_i
- $N_c(f_i)$: Number of classes defined in f_i

—

Complexity Features:

Cyclomatic complexity is defined as:

$$CC(f_i) = 1 + \sum \text{decision points in } f_i \tag{7}$$

where decision points include: `if`, `for`, `while`, `case`, and logical branches.

—

Nesting Depth:

- $ND(f_i)$ measures the maximum depth of nested control structures in f_i

—

Function Length:

$$AFL(f_i) = \frac{1}{N_f(f_i)} \sum_{j=1}^{N_f} L_j \tag{8}$$

where:

- L_j = number of lines in function j
- $N_f(f_i)$ = total number of functions in file f_i

—

IV. Model Design

A. Logistic Regression

$$P(y = 1|x) = \frac{1}{1 + e^{-w^T x}} \tag{9}$$

where:

- x = feature vector of a file
- w = learned weight vector
- $P(y = 1|x)$ = probability that file is defect-prone

—

B. Random Forest

$$\hat{y} = \text{mode}\{T_1(x), T_2(x), ..., T_k(x)\} \tag{10}$$

where:

- $T_k(x)$ = prediction from the kth decision tree
- k = total number of trees in the ensemble
- mode = majority voting across trees

Each tree is trained using:

- bootstrap sampling of training data
- random subset of features at each split

—

C. Model Selection

$$\theta^* = \arg\max_{\theta} F1(\theta) \tag{11}$$

where:

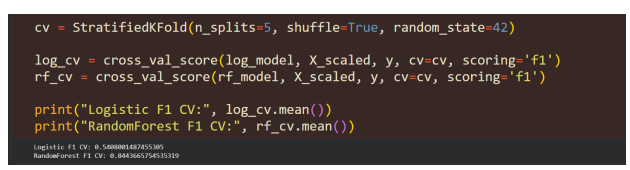
- θ = model hyperparameters
- $F1(\theta)$ = F1-score on validation set
- θ^* = optimal hyperparameter configuration

V. Evaluation Metrics

- Precision: $\frac{TP}{TP+FP}$
- Recall: $\frac{TP}{TP+FN}$
- F1-score
- ROC-AUC

VI. Results

A. Cross-Validation Comparison



```
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

log_cv = cross_val_score(log_model, X_scaled, y, cv=cv, scoring='f1')
rf_cv = cross_val_score(rf_model, X_scaled, y, cv=cv, scoring='f1')

print("Logistic F1 CV:", log_cv.mean())
print("RandomForest F1 CV:", rf_cv.mean())
```

```
Logistic F1 CV: 0.5608001487455305
RandomForest F1 CV: 0.8443665754535319
```

Fig. 2. Cross-validation F1-score comparison between Logistic Regression and Random Forest

Cross-validation results show that Random Forest significantly outperforms Logistic Regression, achieving an F1-score of 0.84 compared to 0.54. This indicates better generalization and ability to capture non-linear patterns.

```
evaluate(log_model, "Logistic Regression")
evaluate(rf_model, "Random Forest")
```

```
Logistic Regression RESULTS
              precision    recall  f1-score   support

           0       0.81      0.91      0.86        46
           1       0.82      0.64      0.72        28

    accuracy                           0.81        74
   macro avg       0.81      0.78      0.79        74
weighted avg       0.81      0.81      0.81        74

Accuracy: 0.8108108108108109
Precision: 0.8181818181818182
Recall: 0.6428571428571429
F1: 0.72
ROC-AUC: 0.8773291925465838
```

Fig. 3. Logistic Regression performance metrics

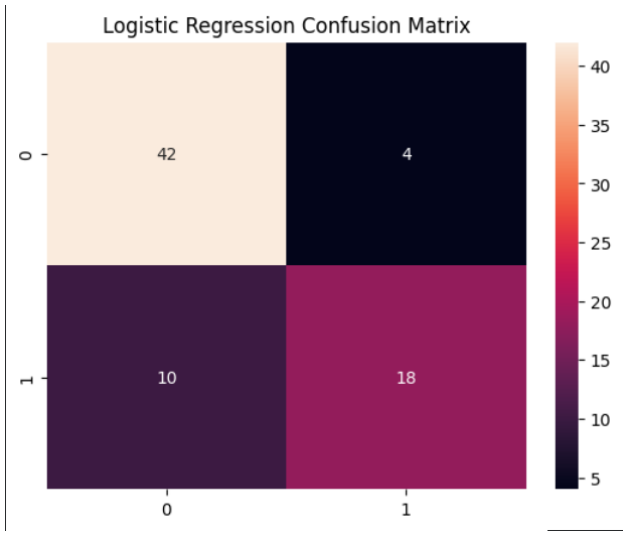


Fig. 4. Confusion Matrix for Logistic Regression

B. Logistic Regression Performance

Logistic Regression achieves moderate performance, with lower recall and F1-score, indicating underfitting due to its linear nature.

C. Random Forest Performance

Random Forest achieves strong predictive performance with:

- Accuracy: 0.89
- Precision: 0.83
- Recall: 0.89
- F1-score: 0.86
- ROC-AUC: 0.90

D. Feature Importance Analysis

Feature importance analysis reveals that:

- Cyclomatic complexity is the most influential feature
- Nesting depth and LOC also contribute significantly
- Function-level metrics provide additional predictive value

VII. Discussion

Results indicate that:

- Random Forest captures non-linear relationships better

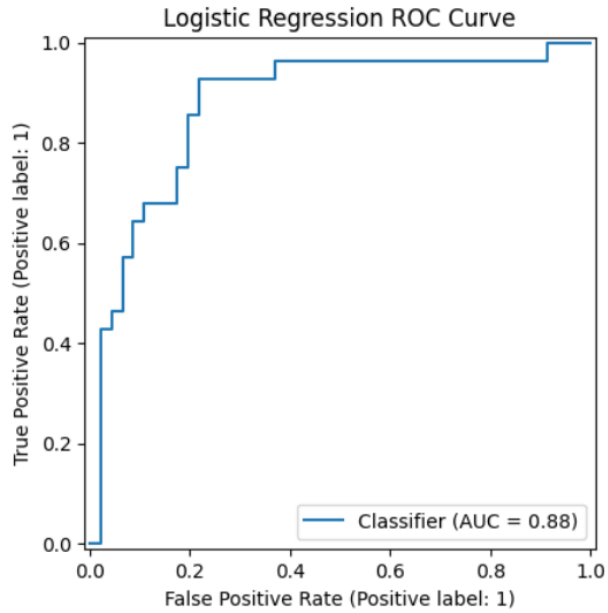


Fig. 5. ROC Curve for Logistic Regression

```
Random Forest RESULTS
              precision    recall  f1-score   support

           0       0.93      0.89      0.91        46
           1       0.83      0.89      0.86        28

    accuracy                           0.89        74
   macro avg       0.88      0.89      0.89        74
weighted avg       0.89      0.89      0.89        74

Accuracy: 0.8918918918918919
Precision: 0.8333333333333334
Recall: 0.8928571428571429
F1: 0.8620689655172413
ROC-AUC: 0.9041149068322981
```

Fig. 6. Random Forest performance metrics

- Complexity and nesting depth are dominant predictors
- Logistic Regression underfits the problem

VIII. System Architecture

The system consists of:

- Frontend: React dashboard
- Backend: Node.js API
- ML Service: FastAPI (Scikit-learn)

A. Project Availability

The implementation of CodeSage is publicly available:

- Source Code Repository: https://github.com/thanos14million605-Internship-2026/CodeSage
- Live Deployment: https://codesage-h1pb.onrender.com

The system provides an end-to-end pipeline including feature extraction, model inference, and a web-based interface for real-time code risk analysis.

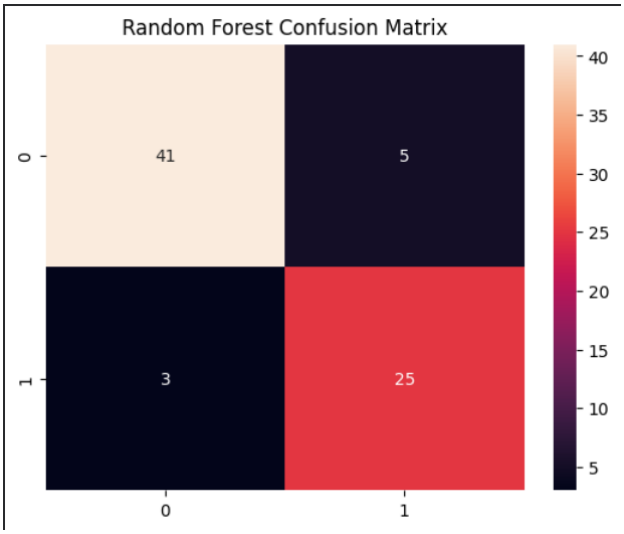


Fig. 7. Confusion Matrix for Random Forest

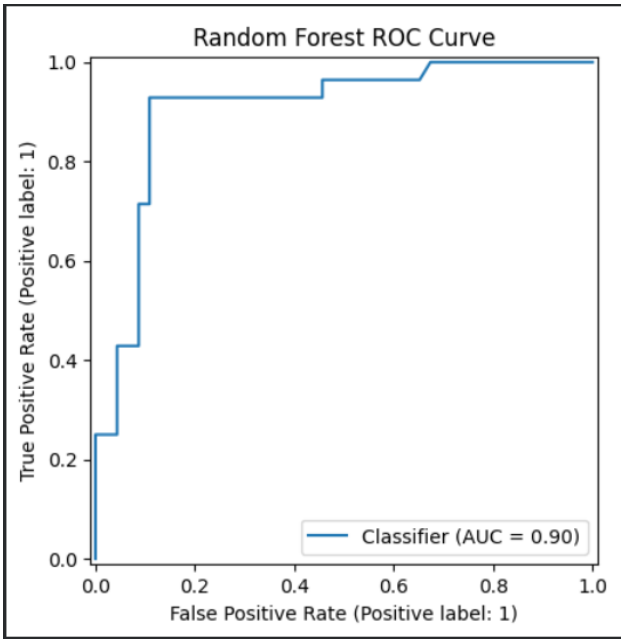


Fig. 8. ROC Curve for Random Forest

IX. Limitations

- Heuristic labels introduce noise
- No real defect ground truth
- Limited to Python codebases

X. Conclusion

Static code metrics can effectively predict defect-prone modules. Random Forest achieves strong performance (F1 = 0.86), demonstrating the predictive power of structural code features.

XI. Future Work

- Incorporate version control features

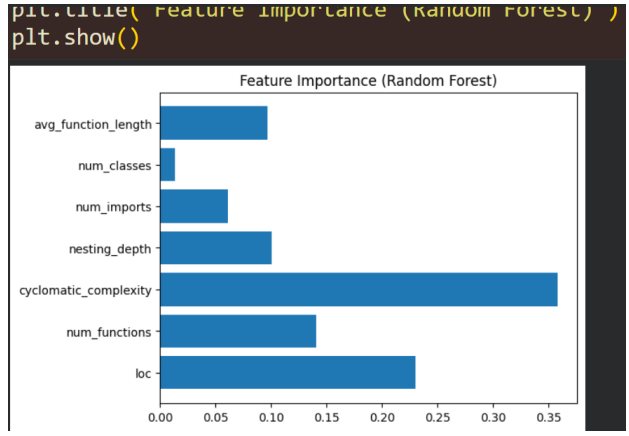


Fig. 9. Feature importance derived from Random Forest

- Use real defect datasets
- Explore deep learning on AST graphs

Acknowledgment

This work was carried out as part of the Bachelor of Computer Applications program at The Maharaja Sayajirao University of Baroda (August 2023 – April 2026), under the guidance of Assistant Professor Pritibala Patel.

References

[1] T. J. McCabe, "A Complexity Measure," *IEEE Transactions on Software Engineering*, 1976.
[2] T. Hall, S. Beecham, and D. Bowes, "A Systematic Literature Review on Fault Prediction Performance," *IEEE Transactions on Software Engineering*, 2012.
[3] S. Lessmann et al., "Benchmarking Classification Models for Software Defect Prediction," *IEEE Transactions on Software Engineering*, 2008.
[4] T. Menzies et al., "Data Mining Static Code Attributes to Learn Defect Predictors," *IEEE Transactions on Software Engineering*, 2007.
[5] N. Nagappan and T. Ball, "Use of Relative Code Churn Measures to Predict System Defect Density," *ICSE*, 2005.
[6] L. Breiman, "Random Forests," *Machine Learning*, 2001.
[7] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, 2011.